

ECO102: Introduction to Macroeconomics

Final Examination, Spring 2026

Duration: 2 hrs | Total marks: 40

Section A – Numerical Problems (20 Marks)

1. Consider the table below representing a hypothetical economy. Assume 2024 is the base year, and the quantities listed for production, imports, exports, and consumption remain constant across both years.

Good	Production	Import	Export	Consumption	Price \$(2024)	Price\$ (2025)
Rice	150	0	0	150	10	12
Milk	100	0	0	100	20	25
Gasoline	0	60	0	60	15	18
Cotton	40	0	40	0	30	32

Given the information above, calculate the following:

- Calculate the Nominal Gross Domestic Product (GDP) for the base year 2024. (2 marks)
- Calculate the GDP Deflator for 2025. (2 marks)
- Calculate the Cost of the Consumer Basket for 2024 and 2025, and determine the Consumer Price Index (CPI) for 2024. (4 marks)
- Calculate the inflation rate between 2024 and 2025 using the CPI. Briefly explain one specific reason why the inflation rate measured by the CPI differs from the inflation rate measured by the GDP Deflator. (2 marks)

2. Consider an economy of people known as the Digitarians. Below is the information for the working-age population and employment status:

Category	Number (millions)
Total working-age population	160
Employed in AI robotics, smart agriculture, virtual design, etc.	108
Unemployed due to automation in retail	5
Recent graduates searching for their first job	5
Unemployed due to recession in manufacturing	4

- Calculate the labor force participation rate. (2 marks)
- Calculate the unemployment rate. (2 marks)
- Calculate the natural rate of unemployment. By comparing your answer to (b), outline the implications for Real GDP, Potential GDP, and the Output Gap. (4 marks)
- Suppose 3 million workers are newly hired, all from the frictionally unemployed group. What percentage of the remaining unemployed population is now structurally unemployed? (2 marks)

Section B – Analytical Problems (20 Marks)

Answer any two questions in this section. When applicable, draw properly labeled diagrams to support your answer.

1. In the recent budget, the Finance Minister of Bangladesh proposed to increase the tax-free income threshold of male taxpayers from Tk 2 lakh 50 thousand to Tk 3 lakh and that of female taxpayers and taxpayers above 65 years of age from Tk 3 lakh to Tk 3 lakh and 50 thousand respectively. Although such measures will help restore the wealth of the public, it also has serious economic implications.
 - a. A reduction in tax will raise the after-tax income and savings. How will it change the condition of the loanable funds market of Bangladesh? Illustrate with an appropriately labelled diagram. (4 marks)
 - b. Do you expect the government to face a budget deficit or a surplus following the tax reduction? How will the condition of the budget affect the loanable funds market of the country? Will the outcome change if the Ricardo-Barro effect is held true? Explain your answer with a relevant graph. (6 marks)
2. The ongoing conflict in the Middle East has led to significant disruptions in global oil supply. As a result, crude oil prices have risen drastically, raising production costs for almost all businesses across the economy. Countries around the world are increasingly concerned about the escalation of this crisis and its potential economic consequences, leading to broad international support for peace talks to restore stability.
 - a. Using the appropriate model(s), explain how this rise in oil prices will affect the economy's real GDP, price level, and unemployment in the short run. Illustrate your answer with an appropriate diagram. (6 marks)
 - b. Based on your answer to part (a), identify the type of output gap that has emerged due to the crisis and explain why this poses a problem for the economy. Discuss how the economy could return to its potential GDP level. (4 marks)
3. In response to persistent inflationary pressures, the central bank of a rapidly growing economy has adopted a contractionary monetary policy. As part of this strategy, it is conducting open-market operations to reduce excess liquidity in the financial system.
 - a. Explain how the central bank carries out open-market operations under a contractionary monetary policy and how these actions affect the supply of money in the economy. Using an appropriate diagram, illustrate and explain the impact on the nominal interest rate. (6 marks)
 - b. Based on your analysis, explain the likely short-run effects of this policy on investment and overall economic activity. (4 marks)

INTRODUCTION TO THE COURSE

Chapter 21: Measuring GDP and Economic Growth

Chapter 22: Monitoring Jobs and Inflation

Chapter 23: Economic Growth

Chapter 27: Aggregate Supply and Aggregate Demand

Mid-Week

Chapter 24: Finance, Savings, and Investment

Chapter 25: Money, The Price Level, and Inflation

Student Presentations (as possible)

Chapter 30: Fiscal Policy

Chapter 31: Monetary Policy